

Technical Document #267: Natural Language Processing - Comprehensive Overview

Technical Document #267: Natural Language Processing - Comprehensive Overview

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Language models predict the probability of a sequence of words.
- Text summarization generates a concise summary of a longer text.
- The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys.